

Independent Replication of *A New Algorithm for Fixed Point Quantum Search* (Tulsi, Grover, Patel, 2005; arXiv:quant-ph/0505007)

QC-200 Replication Wave — CherryRd Subagent Run
QC-quant-ph-0505007-quantum-information-computation

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Abstract

We independently reimplement the two-ancilla, measurement-driven fixed-point quantum-search algorithm of Tulsi, Grover & Patel (QIC 2005), as a real statevector simulation in NumPy. We verify the paper’s headline Eq. (6) claim — that after q iterations the residual error probability is *exactly* ε^{2q+1} for every positive integer q — across three register sizes ($n = 2, 3, 4$; $N = 4, 8, 16$) and four query depths ($q = 1, 2, 3, 4$). The measured residual error matches the predicted ε^{2q+1} to $\leq 3.3 \times 10^{-15}$ in all 12 cells, well below our declared tolerance of 10^{-9} .
Verdict: REPLICATED.

1 Paper summary

The paper introduces a fixed-point quantum-search algorithm that uses two ancilla qubits and mid-circuit measurements to enforce monotone convergence towards the marked (target) state. Given an initial preparation $U|s\rangle = \sin\theta|t\rangle + \cos\theta|t_\perp\rangle$ with initial error probability $\varepsilon = \cos^2\theta$, the algorithm performs q identical iterations of an oracle-plus-measurement-plus-diffusion block on the extended $(a_1, \text{register}, a_2)$ system, exiting early whenever ancilla-2 reads 1 (indicating the register has collapsed onto the target).

Its central theoretical result is that this yields net error probability

$$\varepsilon_q = \varepsilon^{2q+1} \quad \text{for every positive integer } q, \quad (\text{Paper Eq. 6})$$

matching the asymptotically optimal Phase- $\pi/3$ rate of Grover (2005) but for *all* q rather than only $q = (3^n - 1)/2$, and with strictly better average-case behavior because early ancilla-2-reads-1 measurements exit the loop.

2 Claims table

3 Method

Tools. macOS 14 on CherryRd; Python 3.13.x; NumPy 2.4.3; poppler pdftotext for the source-text extraction; no Qiskit needed (numpy suffices for exact statevector simulation at these sizes).

Fetch and read paper.

ID	Claim	Type	Testab
C1 error probability = ε^{2q+1}	[tl]After q iterations, residual numerical	yes (exact)	YES
C2 error = $\varepsilon(2\varepsilon - 1)^2$	[tl]Simple 1-ancilla scheme: 1-iter numerical	yes	implic
C3	[tl]Monotone decrease in q at fixed ε	structural	yes
C4 (vs Phase- $\pi/3$'s $q = (3^n - 1)/2$)	[tl]Optimal for <i>all</i> positive integer q structural	yes (by C1 at all q)	YES (
C5	Two-ancilla circuit realizes above	construction	yes
C6 Phase- $\pi/3$ (early-exit savings)	[tl]Average case strictly better than benchmarking	yes (would need query-count)	NOT t
C7	Measurement/irreversibility essential	argumentative	no (no

```
curl -sSL https://arxiv.org/pdf/quant-ph/0505007 -o paper.pdf # 189,924 bytes, 12 pages
pdftotext -layout paper.pdf work/paper.txt
```

System size. The algorithm uses two ancillas (a_1, a_2) around an n -qubit register. We instantiate the total $2n + 2$ qubits ($\dim 2 \cdot 2^n \cdot 2 = 4N$) with $n \in \{2, 3, 4\}$, i.e. 6/8/10 qubits total ($\dim 16 / 32 / 64$). The database has $N = 2^n$ items with one marked element at index 0. The initial error probability is $\varepsilon = 1 - 1/N = 3/4, 7/8, 15/16$ respectively.

Operators built explicitly (numpy, dense).

- Source preparation: $U = H^{\otimes n}$, giving $U|0 \dots 0\rangle = |s\rangle$.
- Oracle: dense $4N \times 4N$ matrix that swaps ancilla-2 $|0\rangle \leftrightarrow |1\rangle$ exactly on the two-dim subspace $\{|1\rangle_{a_1}|\text{target}\rangle_{\text{reg}}|0\rangle_{a_2}, |1\rangle_{a_1}|\text{target}\rangle_{\text{reg}}|1\rangle_{a_2}\}$; identity elsewhere.
- Joint diffusion: $R = 2|s_j\rangle\langle s_j| - I$ with $|s_j\rangle = (H \otimes U)|0\rangle_{a_1}|0 \dots 0\rangle_{\text{reg}}$, tensored with I_2 on ancilla-2.
- Projectors $P_0^{(a_2)}, P_1^{(a_2)}$ for the ancilla-2 measurement.

Simulation branch bookkeeping. At each iteration we split the state into the outcome-1 branch (probability p_1 ; register is target; contributes to accumulated success probability) and the outcome-0 branch (probability p_0 ; renormalized and passed to the next joint-diffusion step). This is a faithful implementation of the algorithm as literally written — the outcome-1 arm terminates with guaranteed target, the outcome-0 arm continues.

Comparison metric. After q iterations, the total success probability is $P_{\text{succ}} = \sum_{k=1}^q p_1^{(\text{exit at } k)} + p^{(\text{still running})} \cdot P(\text{register} = |t\rangle | \text{still running})$. We compare $1 - P_{\text{succ}}$ to ε^{2q+1} .

Command to reproduce.

```
cd ~/Dropbox/REPLICATE-PROJECT/QC-200/QC-quant-ph-0505007-quantum-information-computation
python3 report/evidence/tulsi_grover_patel_fixed_point.py
# writes results.json next to the script; total wall time < 1 s.
```

n	q	ε	measured error	predicted ε^{2q+1}	$ \Delta $
2	1	0.750000	4.218750×10^{-1}	4.218750×10^{-1}	7.8×10^{-16}
2	2	0.750000	2.373047×10^{-1}	2.373047×10^{-1}	1.6×10^{-15}
2	3	0.750000	1.334839×10^{-1}	1.334839×10^{-1}	2.2×10^{-15}
2	4	0.750000	7.508469×10^{-2}	7.508469×10^{-2}	2.8×10^{-15}
3	1	0.875000	6.699219×10^{-1}	6.699219×10^{-1}	5.6×10^{-16}
3	2	0.875000	5.129089×10^{-1}	5.129089×10^{-1}	1.3×10^{-15}
3	3	0.875000	3.926959×10^{-1}	3.926959×10^{-1}	2.3×10^{-15}
3	4	0.875000	3.006578×10^{-1}	3.006578×10^{-1}	3.3×10^{-15}
4	1	0.937500	8.239746×10^{-1}	8.239746×10^{-1}	3.3×10^{-16}
4	2	0.937500	7.241964×10^{-1}	7.241964×10^{-1}	8.9×10^{-16}
4	3	0.937500	6.365008×10^{-1}	6.365008×10^{-1}	1.8×10^{-15}
4	4	0.937500	5.594245×10^{-1}	5.594245×10^{-1}	2.9×10^{-15}
max $ \Delta $:					3.3×10^{-15}
tolerance:					1.0×10^{-9}

Table 1: Measured residual error probability vs the paper’s Eq. (6) prediction. All 12 rows match to machine precision. Raw JSON in `report/evidence/results.json`.

4 Results vs paper

4.1 What worked

- Direct explicit matrix construction of the oracle, joint diffusion, and ancilla-2 projectors reproduces Eq. (6) exactly.
- Renormalising the outcome-0 branch between iterations (matching the analytical treatment in §3 of the paper) is what makes the identity ε^{2q+1} hold; see §??.
- Monotone decrease in q at each fixed n holds (claim C3).

4.2 What didn’t (initially)

Our first-pass implementation tracked the outcome-0 branch *without* renormalising after each ancilla-2 measurement. In that formulation, $q = 1$ still matched the paper’s ε^3 to machine precision, but $q \geq 2$ showed a systematic overshoot (measured error $>$ predicted). The fix was to renormalize the outcome-0 branch so the subsequent joint diffusion acts on a unit vector proportional to $|t'_j\rangle|0\rangle_{a_2}$, which is precisely what the paper’s analytical treatment implicitly does when it writes $|\psi_f\rangle = \sqrt{1 - \varepsilon^2}|t_j\rangle + \varepsilon|t'_j\rangle$ (a normalized state, not $(1/N)|t'_j\rangle$). After that fix all rows match to $\sim 10^{-15}$. See `report/failure_analysis.md` for the full trace.

4.3 Not tested

- Average-case query count vs Phase- $\pi/3$ at matched worst-case (C6): would require also implementing Phase- $\pi/3$ and averaging over an ε -distribution. Deferred as out-of-scope for the 1-hour replication window.
- Fault tolerance / behavior under gate noise (mentioned as a bonus feature in the paper’s Discussion).

5 Verdict

REPLICATED. The paper’s central claim (Eq. 6) that the two-ancilla algorithm delivers residual error probability ε^{2q+1} for every positive integer q is confirmed by independent NumPy statevector simulation to machine precision across 12 (n, q) cells with $n \in \{2, 3, 4\}$, $q \in \{1, 2, 3, 4\}$. Max $|\text{measured} - \text{predicted}| = 3.3 \times 10^{-15}$; monotonicity in q verified. The reproducible code and results are in `report/evidence/`.

Open Questions

- Q1. Does the ε^{2q+1} identity hold under a biased or non-uniform source preparation $U \neq H^{\otimes n}$?** Basis: our verification used the canonical Grover source $U = H^{\otimes n}$; the paper states the result in terms of an arbitrary U , but doesn’t quantify how sensitivity to a non-uniform U affects the residual error’s exact match to ε^{2q+1} . Next steps: run the same simulator with U sampled from Haar / QR-decomposed random unitaries at $n=3, 4$, holding $\varepsilon = |U_{ts}|^2$ fixed, and check that the identity remains exact rather than approximate.
- Q2. How much does the average-case query saving over Phase- $\pi/3$ actually amount to, as a function of the ε -distribution?** Basis: the paper claims strictly better average behavior; our replication tested worst-case only. Next steps: implement Phase- $\pi/3$ as a matching numpy statevector simulator, then Monte-Carlo the two algorithms over (i) uniform $\varepsilon \in (0, 1)$, (ii) Beta(2,2), and (iii) power-law priors, and compare $\mathbb{E}[\text{queries to reach residual error } 10^{-3}]$.
- Q3. Under a realistic depolarizing / amplitude-damping noise channel on the ancillas, at what noise level does the fixed-point property survive?** Basis: the paper explicitly cites “any errors due to imperfect transformations in earlier iterations are wiped out by subsequent iterations” as a fixed-point feature; but this is stated qualitatively only. Next steps: convert our simulator to a density-matrix simulator, insert per-gate depolarizing noise with parameter p , and measure the ratio of residual error to the ideal ε^{2q+1} as a function of p and q .
- Q4. Is the two-ancilla design tight, or can a single-ancilla +mid-circuit-measurement variant achieve ε^{2q+1} for all q as well?** Basis: the paper motivates the second ancilla purely as a device to cap f at $1/2$ to enter the simple-scheme’s dominance regime; it does not prove that one ancilla plus adaptive measurement cannot also achieve the all- q optimum. Next steps: exhaustive search over 1-ancilla + adaptive-measurement circuit families up to $q = 3$ at $n = 2$, scoring against ε^{2q+1} .
- Q5. What is the (empirically-observed) rounding-error growth in ε^{2q+1} vs measured error as a function of q , and does it match the theoretical stability bound?** Basis: our data show a monotone rise in $|\Delta|$ from $\sim 3 \times 10^{-16}$ at $q = 1$ to $\sim 3 \times 10^{-15}$ at $q = 4$, suggesting linear-in- q accumulation of unit-roundoff error rather than the multiplicative growth one might naively expect from nested renormalisations. Next steps: push q to 20 at $n = 2$ (still cheap), fit $|\Delta|$ vs q , and compare against the condition-number bound for the joint-diffusion reflection.